

MimoApp: A Programming Learning Platform

Applying Programming Models in System Development

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Objectives

MimoApp is an interactive learning platform for programming. This poster highlights the use of various programming models and design patterns applied in developing a scalable, user-friendly system.

- Personalized learning with a focus on object-oriented, event-driven, and functional programming.
- Design patterns applied to create maintainable and scalable features.
- Real-time feedback and assessments for user progress tracking.

Introduction

MimoApp is designed to teach programming languages by providing interactive lessons, real-time feedback, and personalized progress tracking. We integrate Object-Oriented Programming (OOP), Event-Driven Programming (EDP), and Functional Programming (FP) models to ensure a robust learning platform. This project also includes advanced design patterns like Singleton, Factory, and Observer to ensure modularity, scalability, and maintainability.

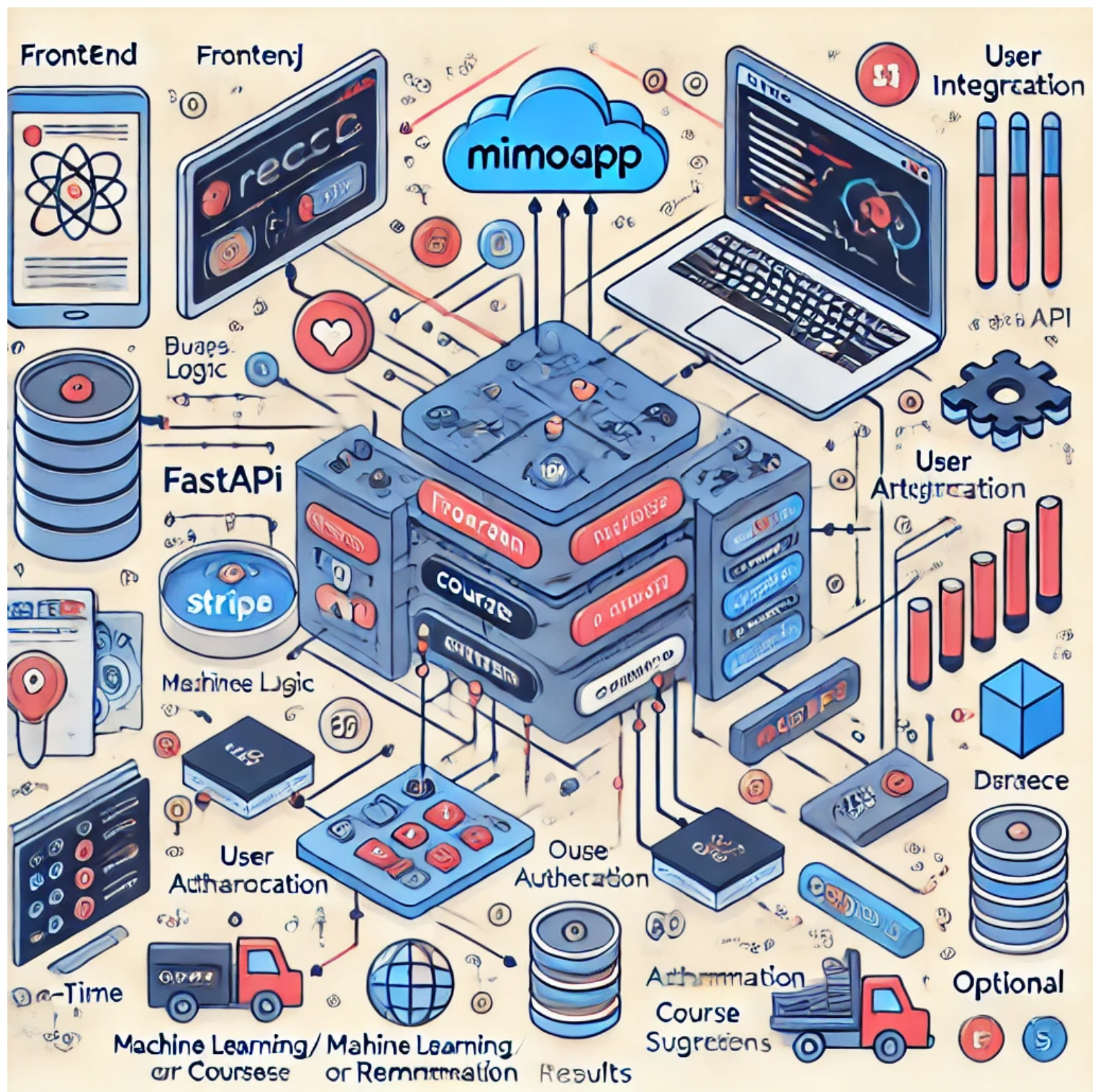


Figure: System architecture diagram of MimoApp.

System Architecture

The system architecture of MimoApp consists of the following layers:

- Frontend: React.js for dynamic interfaces.
- Backend: FastAPI for backend logic and user data management.
- Database: SQL-based system for storing and retrieving content.
- Payment Integration: Stripe for handling premium access to content.

Key Results

The application has shown impressive results in terms of user engagement and system scalability. Users are able to track their learning progress, and the system adapts to their needs through the use of machine learning models and algorithms.

Design Patterns

We utilized several design patterns to enhance modularity:

- **Singleton Pattern:** Ensures a single instance for services like notifications.
- **Factory Pattern:** Simplifies object creation for dynamic user types.
- **Observer Pattern:** Handles real-time updates for users.

Programming Models

The application uses various programming models to achieve flexibility and efficiency:

- **Object-Oriented Programming (OOP):** Organizing entities as objects (e.g., User, Course).
- **Event-Driven Programming (EDP):** Real-time interaction based on user input.
- **Functional Programming (FP):** Ensuring pure functions and immutability.

Conclusion

MimoApp is a scalable and maintainable platform that successfully integrates modern programming models and design patterns. The application has provided valuable insights into learning experiences and is continuously evolving to meet user needs.

Future Work

Future improvements include:

- AI-driven course recommendation system.
- Advanced data analytics for personalized learning.
- Expanded language support for programming tutorials.

Reference

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Results

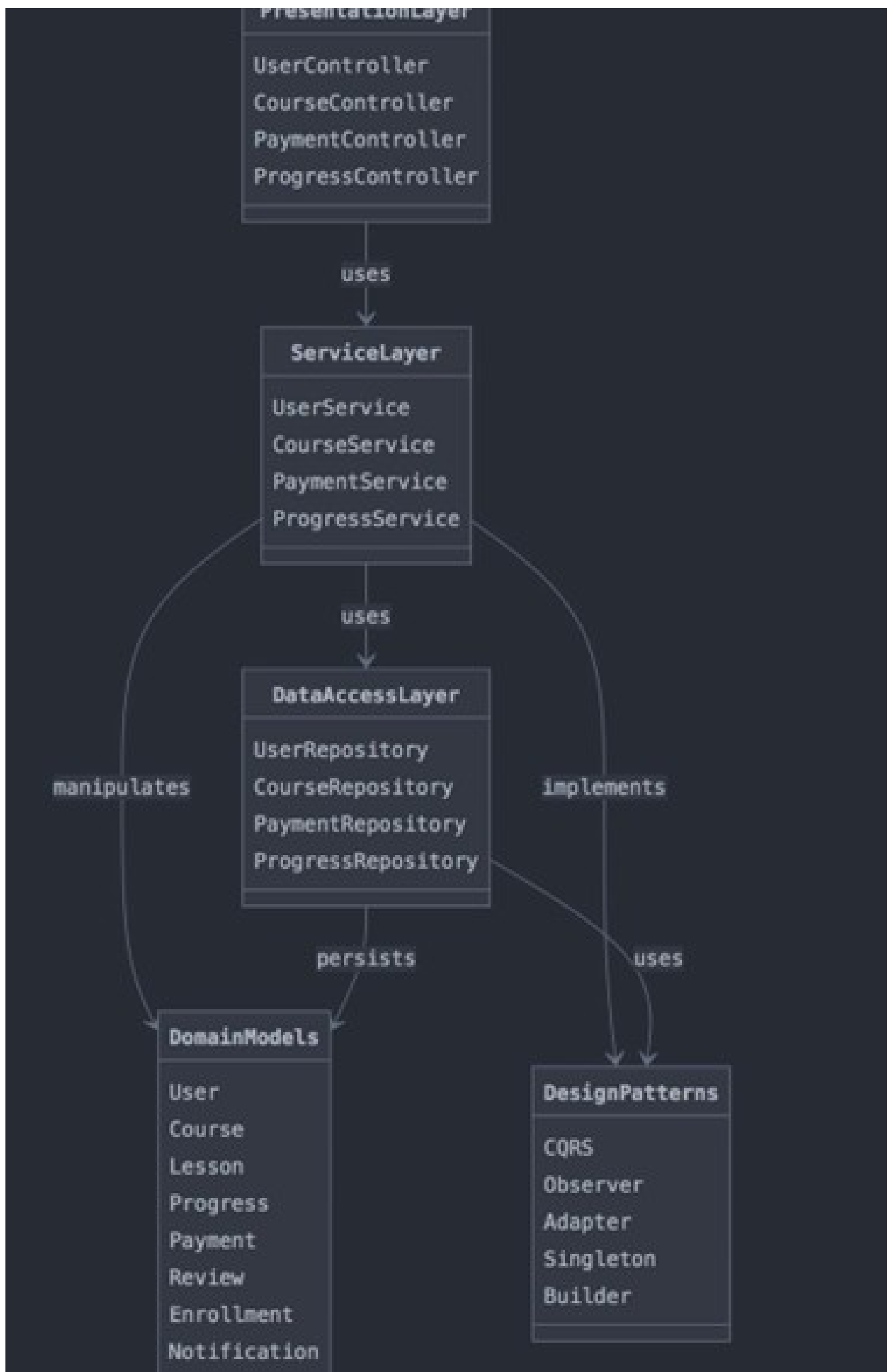


Figure: Results of user engagement and performance improvements.